

EdgeTX Universal Flight Widget Plan

Version: 1.1.00 Date: 23 May 2026 Purpose: Plan for a universal flight widget for TX16S MK3, covering all model types

Hardware Context

Property	Value
Radio	RadioMaster TX16S MK3 / MK3 MAX
Screen	800x480 IPS, landscape
EdgeTX version	2.11+ (LVGL required)
Telemetry sources	RadioMaster sensors + oXs (mstrens, RP2040-based)
Link protocol	TBD per model — assume CRSF/ExpressLRS

Design Philosophy

- Dark background, high-contrast bright text — sunlight readability over aesthetics
 - No bling, no gradients, no decorative elements
 - Large fonts for primary values, smaller for secondary
 - Graceful degradation: if a sensor is absent, that cell is blank or shows "---"
 - Voice announcements handled separately via EdgeTX Special Functions — not in the widget
 - One universal widget, fixed folder name, works across all model types
 - Layout is grid-based with configurable cell positions and font sizes (future-proof)
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Widget Identity

Property	Value
Folder	/WIDGETS/FLTWGT/
File	main.lua
Display name	FltWgt
Screens	2 (swipe between them in EdgeTX main views)

The widget is placed on two separate EdgeTX main views. Each view is a separate widget instance, both using the same `main.lua`, distinguished by an option setting.

Screen 1 — Power

Primary screen. Always the first thing visible when you pick up the radio.

Cell	Value	Font size	Notes
Top-left large	Flight battery voltage	Large	Highest priority
Top-right	Current (Amps)	Medium	Hidden if sensor absent
Bottom-left	Consumed mAh	Medium	Hidden if sensor absent
Bottom-right	(reserved)	—	Future use

Total pack voltage. Per-cell is a post-flight/logging concern, not a during-flight display need.

Screen 2 — Flight Data

Secondary screen. Timer lives here.

Cell	Value	Font size	Notes
Top-left large	Altitude	Large	GPS or baro, whichever is present
Top-right	Speed	Medium	GPS groundspeed or airspeed
Bottom-left	Variometer	Medium	Climb/sink rate — gliders especially
Bottom-right	Timer	Medium	EdgeTX Timer 1

Sensor Handling

The widget reads sensors by name at startup and stores references. If a sensor is not found, the corresponding cell displays --- and does not error.

Sensor names are standard EdgeTX names. oXs will be configured to output these exact names, so no special mapping is needed in the widget.

Data	Sensor name	Notes
Pack voltage (total)	VFAS	Total pack voltage — per-cell not needed during flight
Current	Curr	
Consumed mAh	Capa	
Altitude	Alt	Baro preferred, GPS fallback
Speed	GSpd	GPS groundspeed; airspeed via pitot if available
Variometer	VSpd	Climb/sink rate

Grid Layout System

Each screen is treated as a grid. Cell position and font size are defined in a configuration table at the top of `main.lua`, not hardcoded in the drawing logic.

This means:

- Moving a value to a different position = change one line in the config table
- Changing font size = change one line in the config table
- No hunting through drawing code to adjust layout

Example config structure (not final):

```
lua

local SCREEN1 = {
  { id="volt",   row=1, col=1, span=2, font=DBLSIZE, label="V"
    { id="curr",   row=1, col=3, span=1, font=MIDSIZE, label="A"
      { id="capa",   row=2, col=1, span=1, font=MIDSIZE, label="mAh"
    }
  }
}
```

Font sizes available in EdgeTX Lua: `SMSIZE`, `STDSIZE`, `MIDSIZE`, `DBLSIZE`, `XXLSIZE`

Grid dimensions and cell pixel sizes will be calculated from `zone.w` and `zone.h` multiplied by `lvgl.LCD_SCALE` — no hardcoded pixel values.

What We Still Need to Discuss (per screen, per value)

Before writing layout code, for each cell we need to agree on:

1. Font size
2. Show label/unit alongside value, or not (e.g. "12.4V" vs just "12.4")
3. Color coding — e.g. voltage turns red below threshold
4. Warning thresholds — at what voltage/mAh do we want visual alert
5. Number of decimal places per value
6. Whether per-cell average voltage is shown instead of (or alongside) pack total

This is a short conversation per screen, not a long one.

Development Sequence

Step 1 — Skeleton widget

- LVGL structure, two-screen option, grid renderer, no real sensors
- Displays static placeholder values
- Confirms layout and font sizes look right on hardware

Step 2 — Sensor wiring

- Connect real sensor reads
- Graceful handling of absent sensors
- Test with actual oXs output

Step 3 — Visual refinement

- Adjust font sizes, spacing, color thresholds based on what you see on hardware
- Sunlight readability check

Step 4 — Config table

- Move all layout parameters into config table at top of file
- Verify moving a cell or changing font size is a one-line change

Step 5 — Field test

- Fly, check readability and usefulness
- Adjust thresholds and layout based on actual use

Out of Scope (for now)

- Voice announcements — handled by EdgeTX Special Functions, not the widget
- Link quality display
- Third screen
- Model-specific configuration files
- oXs configuration — separate project, must align sensor names with EdgeTX expectations

Reference

- LVGL API: <https://luadoc.edgetx.org/lua-api-reference/lvgl-for-lua/api>
- Widget Scripts overview: <https://luadoc.edgetx.org/overview/script-types/widget-scripts>
- oXs project: https://github.com/mstrens/oXs_on_RP2040
- EdgeTX lua-scripts gallery: <https://github.com/EdgeTX/lua-scripts>